Subextensions of Galois extensions

M/K : Galois with Galois gp $G = \mathrm{Gal}(M/K)$

Proposition (1):

$$\begin{array}{c} M \\ | \\ L \\ | \\ K \end{array} \leftarrow \text{sub-extension}$$

Then:

(1) M/L is also Galois

(2) $H = \mathrm{Gal}(M/L) = \{\sigma \in G : \sigma|_L = \mathrm{id}\} \leq \mathrm{Gal}(M/K) = G$

(3) $\mathrm{Gal}(M/K)/\mathrm{Gal}(M/L) = G/H \xrightarrow{\cong} \mathrm{Emb}_K(L, M)$

$$\sigma \longmapsto \sigma|_L$$

Pf: (1) Can use the splitting field characterization of Galois extn

(2) Easy

homomorphism of group actions

(3)
$$\begin{tikzcd} G \curvearrowright G \arrow[d] \arrow[dr, "\sigma \mapsto \sigma|_L"] & \\ G \curvearrowright G/H \arrow[r, "\cong"] & G \curvearrowright \underbrace{\mathrm{Emb}_K(L, M)} \end{tikzcd}$$

$(\sigma_1, \sigma_2 H) \mapsto \sigma_1 \sigma_2 H$

$\sigma \in G$

$$\begin{tikzcd} L \arrow[r, "\lambda"] \arrow[dr, "\sigma \circ \lambda"'] & M \arrow[d, "\sigma"] \\ & M \end{tikzcd}$$

$(\sigma, \lambda) \mapsto \sigma \circ \lambda$

H = Stabilizer of identity embedding of $L \longrightarrow M$

Proposition (2): $\sigma \in G, \ \sigma(L) \leq M$

$$\begin{tikzcd}[column sep=small] & M \arrow[dl, no head] \arrow[dr, no head] & \\ L \arrow[dr, no head] & & \sigma(L) \arrow[dl, no head] \\ & K & \end{tikzcd}$$

(1) $Gal(M/\sigma(L))$
$= \sigma\, Gal(M/L)\, \sigma^{-1}$
$\leq Gal(M/K)$

(2) TFAE:

(a) $Gal(M/L) \trianglelefteq Gal(M/K)$
normal

(b) $L = \sigma(L), \ \forall\, \sigma \in Gal(M/K)$

(c) L/K is Galois

Pf: (1) is on HW 10 (problem 5)

(2) (b) $\Rightarrow$ (a) follows from (1)

(2) (b) $\Leftrightarrow$ (c)

L/K is Galois

$\Leftrightarrow |Emb_K(L, L)| = [L:K]$

$$\begin{gathered} \shortparallel \\ |\mathrm{Emb}_K(L,M)| \end{gathered}$$

$\Longleftrightarrow \mathrm{Emb}_K(L,L) = \mathrm{Emb}_K(L,M)$

$\Longleftrightarrow \forall \sigma \in \mathrm{Gal}(M/K)$

$(\sigma|_L)(L) \subseteq L$

$\Longleftrightarrow \forall \sigma \in \mathrm{Gal}(M/K),$

By dimension reasons $\quad \sigma(L) = L$

To see (a) $\Rightarrow$ (b), we will need to know <u>more</u>, namely the Corollary below.

$$\begin{gathered} \mathrm{Gal}(M/\sigma(L)) \\ \shortparallel \\ \mathrm{Gal}(M/L) \text{ normal } \Rightarrow \sigma\, \mathrm{Gal}(M/L)\, \sigma^{-1} \\ \shortparallel \qquad \forall \sigma \in G \\ \mathrm{Gal}(M/L) \end{gathered}$$

$\Rightarrow \sigma(L) = L, \forall \sigma \in G$

Rmk: We have constructed

$$\left\{ \begin{array}{c} L|K : \text{subextensions} \\ \text{of } M|K \end{array} \right\}$$

$$\downarrow \qquad \begin{array}{c} L|K \\ \downarrow \\ \mathrm{Gal}(M/L) \end{array}$$

$$\left\{ \begin{array}{c} \text{subgroups} \\ \text{of } \underbrace{\mathrm{Gal}(M/K)}_{G} \end{array} \right\} \quad \leq \mathrm{Gal}(M/K)$$

Defn $H \leq G = \mathrm{Gal}(M/K)$

The fixed field of H is

$$M^H = \{ \alpha \in M : \sigma(\alpha) = \alpha \ \forall \sigma \in H \}$$

Observation: M^H/K is a subextension

Pf: $K \leq M^H$ & M^H is closed under

$+, \cdot, (\,)^{-1}$.

Rmk (2)

$$L/K \qquad\qquad \left\{\begin{array}{c}\text{subextensions} \\ \text{of } M/K\end{array}\right\}$$

$$\mapsto \qquad\qquad\qquad \downarrow$$

$$\mathrm{Gal}(M/L) \qquad\qquad \left\{\begin{array}{c}\text{subgroups of} \\ G = \mathrm{Gal}(M/K)\end{array}\right\}$$

$$H$$

$$\mapsto \qquad\qquad \downarrow$$

$$M^H \qquad \left\{\begin{array}{c}\text{subextensions} \\ \text{of } M/K\end{array}\right\}$$

Thm (Main thm of Galois theory)

These assignments are inverses to each other. That is:

(1) $$L = M^{\mathrm{Gal}(M/L)}$$

(2) $H = \mathrm{Gal}(M/M^H)$

Corollary: The assignment

$$L|K \longmapsto \mathrm{Gal}(M/L)$$

is a bijection

$$\left\{ \begin{array}{c} \text{Subextensions} \\ L|K \\ \text{of } M/K \end{array} \right\} \xrightarrow{\cong} \left\{ \begin{array}{c} \text{Subgroups} \\ H \leq G \end{array} \right\}$$

Observation (2)

We always have inclusions

- $L \leq M^{\mathrm{Gal}(M/L)}$
- $H \leq \mathrm{Gal}(M/M^H)$

$$\begin{gathered} \| \\ \{\sigma \in G : \sigma|_{M^H} = \mathrm{id}\} \end{gathered}$$

Need to check that these two inclusions are bijections.

<u>Proposition :</u> M/K Galois

$$K = M^{\mathrm{Gal}(M/K)}$$

<u>Pf:</u>

$$\begin{array}{c} M \\ \Big| \, \Big\} [M:L] \\ L = M^{\mathrm{Gal}(M/K)} \\ \Big| \, \Big\} [L:K] \\ K \end{array}$$

$$|\mathrm{Gal}(M/L)| = [M:L]$$

But $\mathrm{Gal}(M/K)$ fixes L

$$\Rightarrow \quad |\mathrm{Gal}(M/L)| \geq |\mathrm{Gal}(M/K)|$$

$$\Rightarrow \quad [M:L] = [M:K]$$

$\Rightarrow [L:K] = 1$

$\Rightarrow K = L$ ✓